



Inequality notation to express error

- 1 An error interval for a number x shows the range of possible values of x . It is written as:
Tick **1** correct answer

☐ $a < x < b$

☐ $a > x < b$

☐ $a \geq x < b$

☒ $a \leq x < b$

- 2 When a number is rounded to the nearest integer the result is 8. Identify the correct range for the unrounded number. Tick **1** correct answer

☒ Smallest = 7.5, Largest = $8.\dot{4}9$

☐ Smallest = 7.5, Largest = $8.9\dot{9}$

☐ Smallest = 7.5, Largest = $8.5\dot{9}$

☐ Smallest = 7.5, Largest = 8.5

- 3 When a number is rounded to the nearest integer the result is 19. Identify the correct range for the unrounded number. Tick **1** correct answer

☐ Smallest = 18.5, Largest = $19.5\dot{9}$

☒ Smallest = 18.5, Largest = $19.4\dot{9}$

☐ Smallest = 17.5, Largest = 19.5

☐ Smallest = 18.5, Largest = 19.5



- 4 When a number, x , is rounded to the nearest integer the result is 8. Identify the error interval for x . Tick **1** correct answer

☐ $7.5 < x < 8.4\dot{9}$

☐ $7.5 > x < 8.4\dot{9}$

☐ $7.5 \leq x < 8.4\dot{9}$

☒ $7.5 \leq x < 8.5$

☐ $7.5 \geq x < 8.4\dot{9}$

- 5 When a number, x , is rounded to 2 significant figures the result is 7.8. Identify the error interval for x . Tick **1** correct answer

☐ $7.75 < x < 7.84\dot{9}$

☒ $7.75 \leq x < 7.85$

☐ $7.74\dot{9} < x < 7.84\dot{9}$

☐ $7.75 < x < 7.85$

- 6 A distance x km is given as 2.345 km correct to the nearest metre. Identify the error interval for x . Tick **1** correct answer

☐ $2.3445 \leq x < 2.3449$

☐ $2.3445 \geq x < 2.3454\dot{9}$

☐ $2.3445 > x < 2.3454\dot{9}$

☐ $2.3445 < x < 2.3455$

☒ $2.3445 \leq x < 2.3455$

